

Safe System of Work

Use of Mobile Crusher

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Overview

The following information provides guidance on the company's Standard Operation Procedures for works involving the use of Mobile Crusher

All employees have a responsibility for their safety and the safety of others by adhering to these codes of practice.

Mobile crushers are used in a wide range of industries, including quarrying, ore processing and recycling of demolition waste. This guidance is for all those with responsibilities for the operation of mobile crushers, including contract managers, supervisory staff and operators of these machines. It covers the safe operation of the mobile crushing operation and includes guidance on clearing blocked crushers.

Mobile crushers are versatile pieces of equipment with many uses. A properly planned operation will be efficient, safe and productive. A poorly planned operation will be unproductive, dangerous and inefficient.

The risks associated with your particular operation, and the methods of reducing those risks, should be revealed during your risk assessment.

These machines can, and have, killed.

Hazards Identified

The use of a Crushing Plant could result in crushing injuries, collisions with people or vehicles or being struck by debris, these can all result in serious injury or can possibly be fatal.

- Adverse weather
- Machine Blockages
- Crushing
- Cuts and abrasions
- Vehicle movements

- Falls from height
- Slips & Trips

Personal Protective Equipment (PPE) Requirements



Operations

Typical layouts

Mobile crushers will normally be fed from loading shovels, backhoe excavators or from other processing machinery (e.g., sizer/separator, elevating conveyor, etc).

In its simplest form it will be a standalone machine, fed directly with the material to be crushed and discharging to a stockpile (see Fig 1).

If the material to be crushed is reinforced concrete, then generally a magnetic separator will often be positioned over the discharge conveyor to remove the metal and prevent it from contaminating the production run material.



Fig. 1 Typical mobile stone crusher

Feeding

If the mobile crusher is to be fed directly by a loading shovel or excavator, then:

- Excavator standing pads should be suitable (stable) and should be high enough for the operator to be able to monitor the feed hopper from the cab. Safe access to the excavator must be provided for the operator (see Fig 2).
- Where wheeled loading shovels are used, then the ramp should be wide enough to allow for adequate edge protection (min 1.5m high) on either side of the ramp as well as for the travel of the machine.

- The maximum gradient of the ramp should be within the capacity of the loader (a maximum gradient of 1:10 is recommended)
- The last few metres of the ramp should be level so that the machine is not discharging uphill, thus enabling operators to monitor the feed more easily. The loading shovel will also be more stable.
- Ensure pedestrians and obstructions are excluded from the bucket operating arc.

If the crusher is to be fed directly by a conveyor:

- All dangerous parts of machinery should be suitably guarded.
- Failure to replace guards (on crushers and conveyors) after cleaning or maintenance work is a major cause of injury on these machines.



Fig. 2 Excavator feeding mobile stone crusher

Blocked crushers

Causes

Causes of crusher blockages can be grouped under two main headings:

- Stalling due to:
 - Electrical or mechanical failure
 - Material jammed in the chamber causing an overload
 - Overfeeding material
 - Entry of tramp metal or wood
 - Accumulation of material in the crash box

- Accumulation of fine material in the crusher discharge chute
- Bridging due to:
 - Oversize feed material
 - Excessive clay or other fines in the crushing cavity preventing small material passing through the crusher
 - A foreign body in the crusher feed or discharge chamber obstructing the feed material

Prevention

Working on the premise that prevention is better than cure, every effort should be made to prevent oversize material or tramp metal entering into the crusher feed hopper by:

- Designing any quarry blast to achieve optimum rock fragmentation (quarrying)
- Reducing oversize material by drop ball or hydraulic hammer or other means (demolition/construction)
- Training and instructing the loader driver not to load oversize material
- Sizing bars on crusher feeds
- Following the manufacturer's recommendations on the rate, presentation of feed and crusher settings
- Instituting a programme of good housekeeping to prevent scrap steel entering into shovel buckets
- Ensuring the size of buckets are appropriate to the capacity of the crusher
- Regular inspection of metal parts (e.g. bucket teeth, dumper wear plates and drilling components etc) to ensure they are unlikely to break off and enter the crusher feed
- The strategic placing of electrical magnets and/or the installation of metal detectors to prevent tramp metal from entering the crusher
- The use of level indicators for feed control
- Maintenance of drive systems
- Removal and adequate cleaning of the discharge chute

A properly designed mobile crushing operation should not need any person to be present on the crusher access platform during normal crushing operations.

Being on the access platform during normal operation presents the following risks:

- Struck by objects ejected from the crusher, such as bits of stone or metal.
- The operation of the crusher catches being struck by any reinforcing bar as it.
- Being pulled into the crushing chamber when attempting to pull out contaminants (e.g. reinforcing bar).

- Struck by the loader bucket if the access platform is within the working radius of the loading machine (for 360 degree excavators).
- Falling if adequate guardrails and access arrangements are not provided.
- Noise. Process noise at this level can cause deafness and adequate hearing protection is required.
- Dust, especially in the case of crushing concrete or bricks or any other high silica content material. Breathing silica dust can cause serious respiratory diseases. [HSE Guidance on Silica](#).
- Whole body vibration. Anyone on the platform would be subjected to constant low frequency vibration.

It may be necessary for a person equipped with the appropriate PPE (e.g. ear defenders, dust mask, eye protection, hard hat, protective footwear, high visibility outer garment) to spend a few minutes setting the feed speed initially if there is no remote facility. The feed should then be controlled from the machine feeding the crusher by varying the loading rate into the feed hopper.

Action when a crusher becomes blocked

- Have a nominated person to supervise the activity and call for assistance as necessary
- Stop the feed at the earliest opportunity
- Remove excess material by mechanical means where possible before the cause of the blockage can be dealt with
- In some cases, however, an amount of removal by hand will be involved and when this occurs the crusher and associated plant must be stopped and isolated
- Manual removal should only be carried out by suitably trained and competent persons

Hazards encountered may be:

- Poor or difficult access
- Accidental start-up of feeder, crusher or adjacent plant
- Being struck by material from the feeder, chute or projected material
- The movement of any material present inside the crusher
- Slipping and falling
- Manual handling of rocks and equipment
- Unexpected movement of crusher components
- Damaged electrical cable
- Noise
- Stored energy from electrical, hydraulic, compressed air, mechanical sources and gravity
- Unsafe placement of material removed from the crusher.

Clearing blockages

Bridged crushers

The preferred method of clearing a bridged crusher is by the use of a hydraulic arm (typically a360 excavator fitted with a quick hitch bucket attachment and either a static pick or a hydraulic hammer available).

Depending on the result of your risk assessment, clearing out a bridging blockage with a hydraulic arm (with pick attachment) may be carried out with or without the crusher still operating.

Use of cranes (including manually operated lifting tackle), Lifting and Slinging

When hydraulic arms are not available and cranes slung with hooks have proved ineffectual, other methods are necessary. When it becomes necessary for a person to enter the crusher to position hooks or slings, the crusher and feeder must be stopped, isolated and locked off (in accordance with manufacturers/suppliers' instructions) and safety harness worn.

Other available options

Where other options are to be considered they should be subject to a detailed and thorough risk assessment. The crusher should be shut off and isolated before considering the use of bars and hand hammers. Careful consideration should be given to the risk of large pieces of feed material moving and causing trap or crush injuries. Wedges should not be used due to the risk of them becoming a projectile (This has caused fatalities in the past). Other options, which require more specialist expertise and competence, include gas or chemical expansion, hydraulic ramp plates, permanently mounted hydraulic breaker (may include the use of closed circuit television CCTV to assist the operator), use of explosives.

Stalled crushers

A stalled crusher should be treated as possibly being jammed with tramp metal, which could be ejected with fatal consequences. Written instructions should be issued to plant operators. This should detail the procedures to follow in the event of a crusher stalling. These instructions should include the following:

- Isolation of motive power to the crusher and associated plant procedure in operation at the site
- Clear the area of all personnel
- Notify the site manager of the stalled crusher

If, after careful examination, there appears to be no electrical or mechanical reason why the crusher has stalled, it may indicate that the crusher is jammed by tramp metal.

Wherever possible any inspection of the crushing cavity of a jaw crusher should be carried out from below the crusher, not from above.

Remember: -

Fatal accidents (due to material being ejected) have occurred to people who have examined the crushing cavity of a stalled crusher from above.

Guarding

Inadequate guarding is a major cause of injury. Guarding integrity should be checked at regular intervals, particularly after cleaning or maintenance work.

Slips and Trips - spillage

Keeping clear working areas clear of spillages improves operational efficiency and also reduces the risk of slips and trips – a major cause of lost time accidents.

Minimise spillages by:

- Maintaining conveyor skirts
- Maintaining the correct adjustment and condition of belt scrapers. Ensure that any feed conveyor discharges centrally into the feed hopper.
- Designing loading shovel ramps to minimise the amount of material that will spill from the ramp itself.

Never remove guards to clean up while the machine is in operation. If guards need to be removed, shut off **and** lock off.

Transport and Segregation

To minimise the transport risks on site, a risk assessment should address:

- Safe site
- Safe equipment
- Safe worker

It is essential where possible to segregate pedestrians from moving plant such as loading shovels, excavators and dump trucks by the use of physical barriers.